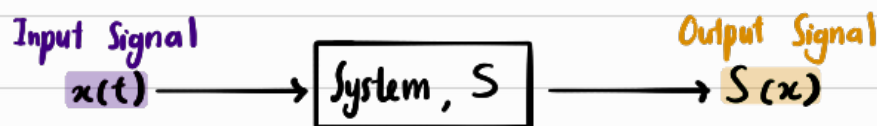


Systems [Function of Functions]

A system is a function of a function. It takes an input signal function and transforms it into an output signal function.



Mathematically, $y = S(x)$ OR $y = Sx$
Function notation Matrix-vector multiplication notation

Important Notations

Assume x_i are real valued signals
 t is one particular time
 S is a system

x_i : Entire Signal / Function

$x_i(t)$: Signal value of x_i at time t

$S(x_i)$: Entire Output Signal

$[S(x_i)](t)$ OR $Sx_i(t)$: Value of the Entire Output Signal at time, t
[Evaluates the Entire Output Signal at single moment, t]

$S(x_i(t))$: Value of system acting on the singular signal at time, t
[Evaluates input signal at time t b4 any system processing happens]

Simple Systems

① Adder

An Adder takes input signals & produces their point-by-point sum



$$y = x_1 + x_2 + x_3 + \dots$$

② Gain / Scaling System

A gain system multiplies the signal value by a constant K :



$$y = Kx$$

③ Full-wave Rectifier

A full-wave rectifier outputs the magnitude of the input signal



$$y = |x|$$

④ Delay System

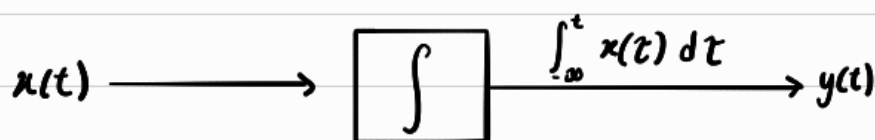
A Delay moves the entire signal later in time



$$y(t) = \text{Delay}_\tau(x)(t) = x(t - \tau)$$

⑤ Integrator

The integrator accumulates the input signal from the past up to the present time.

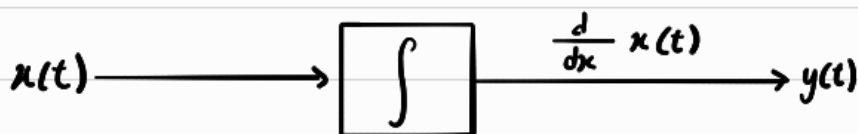


$$y(t) = \int_{-\infty}^t x(\tau) d\tau$$

Dummy Variable

⑥ Differentiator

A differentiator returns the instantaneous rate of change of the input



$$y(t) = \frac{d}{dt} x(t)$$

Example Question

Signals x and y are the input & output of an **integrator system**

$$x(t) = e^{-t} \overset{\text{Unit Step}}{u(t)} \quad \text{for all } t$$

Find an expression for $\text{Delay}_1(y)(t)$ for all t



$$u(t) \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases} \implies x(t) \begin{cases} 0, & t < 0 \\ e^{-t}, & t \geq 0 \end{cases}$$

Since $x(t)$ went through an integrator system,

$$\begin{aligned} \implies y(t) &= \int_{-\infty}^t x(\tau) d\tau \\ &= \int_{-\infty}^0 0 d\tau + \int_0^t e^{-\tau} d\tau \\ &= [-e^{-\tau}]_0^t \end{aligned}$$

$$y(t) = 1 - e^{-t}$$

Integrator System

$$\therefore y(t) = \begin{cases} 0, & t < 0 \\ 1 - e^{-t}, & t \geq 0 \end{cases} \implies y(t) = (1 - e^{-t}) u(t)$$

$$\text{Let } z = \text{Delay}_1(y)(t)$$

$$= y(t - 1)$$

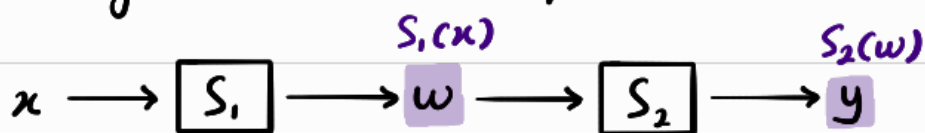
$$z = [1 - e^{-(t-1)}] u(t-1) \quad \#$$

Delay System

Interconnecting / Composite Systems [System of Systems]

① Series / Cascade

The output of one system becomes the input of the next



$$y = S_2(S_1(x)) \xrightarrow{\text{Abbreviated}} y = S_2 S_1 x$$

Compatibility Requirement

The output produced by S_1 must be a valid input for S_2

Example Question

Consider the series interconnection of discrete-time system



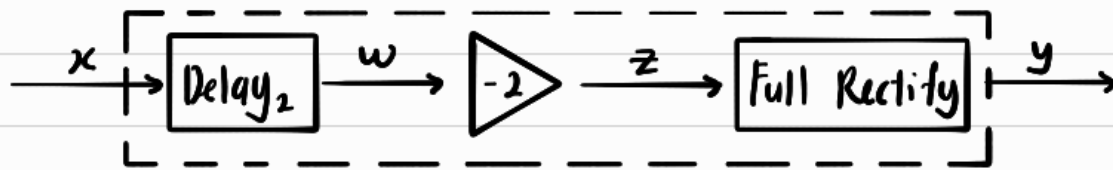
Find a formula for the output y at time n

$$\begin{aligned} \text{Delay} &: w[n] = x[n-2] \\ \text{Rectifier} &: z[n] = |w[n]| \\ \text{Gain} &: y[n] = -2z[n] \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{Delay} \\ \text{Rectifier} \end{aligned}} \right\} z[n] = |x[n-2]| \quad \text{--- ①}$$
$$\text{Gain} : y[n] = -2z[n] \quad \text{--- ②}$$

$$\text{①} \rightarrow \text{②} : y[n] = -2|x[n-2]| \quad \#$$

Example Question [Changing Order]

Consider the series interconnection of discrete-time system



Find a formula for the output y at time n

$$\left. \begin{array}{l} \text{Delay} : w[n] = x[n-2] \\ \text{Rectifier} : z[n] = -2w[n] \end{array} \right\} z[n] = -2x[n-2] \text{ --- ①}$$

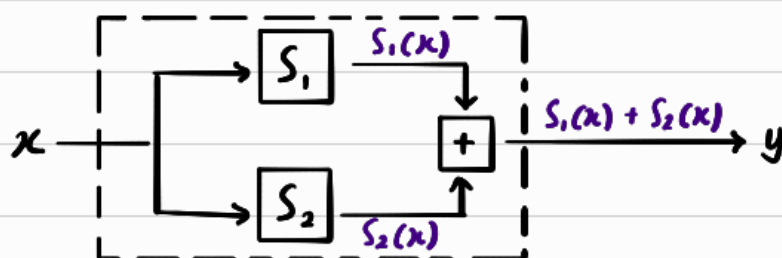
$$\text{Gain} : y[n] = |z[n]| \text{ --- ②}$$

$$\text{①} \rightarrow \text{②} : y[n] = |-2x[n-2]| = 2|x[n-2]| \quad \#$$

Note that negative gain & full rectifier do not commute

② Parallel

The same input is sent to multiple systems.

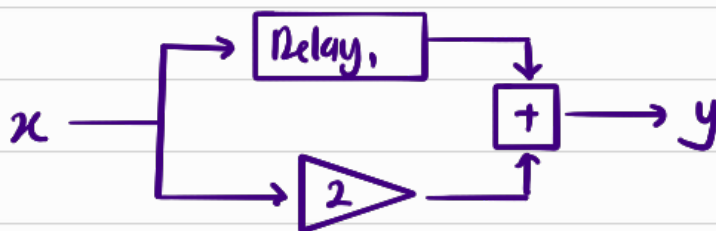


$$y = S_1(x) + S_2(x)$$

Example Question

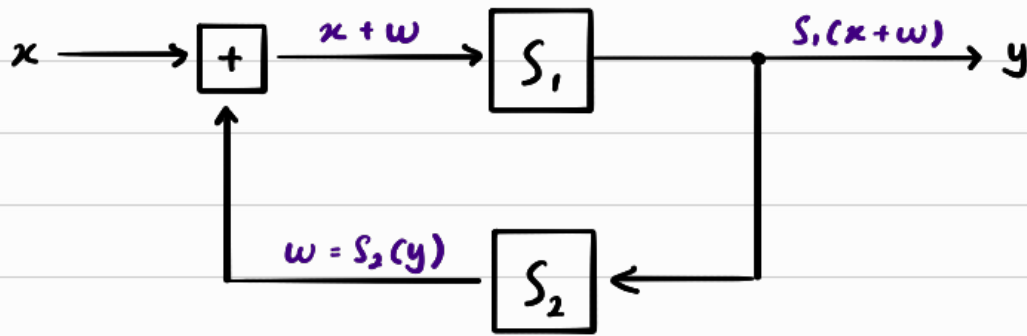
Draw a block diagram representing the following system

$$y[n] = \underbrace{x[n-1]}_{\text{Delay,}} \underbrace{+}_{\text{Parallel}} \underbrace{2x[n]}_{\text{Gain = 2}}$$



③ Feedback

The output is processed & "fed back" to an earlier part of the system



$$y = S_1(x + w)$$

↓

$$y = S_1(x + S_2(y))$$

Note that the Eqn contains "y" on both sides

System Properties

Which input timeline does the output depend on?

① Memoryless

Output at a particular time ONLY depends on the input at that time

Essentially, output now only uses input now.

Continuous - time : $y(t) = F(t, x(t))$

Discrete - time : $y[n] = F(n, x[n])$

With memory

Output now require input at some other time.

Eg. $y[n] = \frac{1}{2} [x[n] + x[n-1]]$ OR $y(t) = \frac{d}{dx} x(t) - x(0)$

② Causal

Output now only depends on input before OR now [not future]

$y[n] = \frac{1}{2} [\overset{\text{present}}{x[n]} + \overset{\text{past}}{x[n-1]}]$; $y[n] = \frac{1}{2} x[n]$; $y(t) = \int_{-\infty}^t e^{x(\tau)} d\tau$
[Memory, causal system] [Memoryless, causal] [Memory, causal]

Take Note :

Every memoryless system is causal but NOT every causal system is a memoryless system

A causal system just have to have an output that doesn't depend on the future.

Stability

① Bounded Input Bounded Output (BIBO)

Every bounded input produces a bounded output

$$\text{Bounded Input} \\ |x(t)| \leq 1 \text{ for all } t$$

$$\underbrace{|x(t)|}_{\text{finite}} \leq \underbrace{M_x}_{\text{finite}} \implies |y(t)| \leq M_y \text{ for all } t \\ -\infty \leq t \leq \infty$$

To prove a system is stable, we must show

Every bounded input produces a bounded output

To prove a system is unstable, we must show

One bounded input $\rightarrow$ unbounded output

Example 1

Is $y(t) = |x(t)|$ BIBO stable?

Assume input signal bounded
 $\implies |x(t)| \leq M$

$$\begin{aligned} \implies |y(t)| &= ||x(t)|| = |x(t)| \\ \implies |y(t)| &= |x(t)| \end{aligned}$$

Since $y(t) = |x(t)|$

$\therefore |y(t)| \leq M$. BIBO stable

Example 2

Is $y(t) = \int_{-\infty}^t x(\tau) d\tau$ BIBO stable?

Let's try to prove its BIBO unstable

Unit Step
Function

Let bounded input signal, $x(t) = u(t) \leq 1$

$$\therefore y(t) = 0 + \int_0^t 1 d\tau$$

$$y(t) = t$$

As $t \rightarrow \infty$, $y(t) \rightarrow \infty$

$\therefore$ We have proven for a given bounded input, we got a unbounded output
 $\implies$ BIBO unstable.

Time-Invariant (TI) Systems

System behaviour does not depend on absolute time.

$$x(t - \tau) \xrightarrow{S} y(t - \tau)$$

If system, S is time-invariant, the only effect should be that its output is delayed by the same amount. Nothing about the output's shape or behaviour should otherwise change.

Formal proof for TI system

The system must commute with delay

System then Delay
Delay then System

$$\begin{array}{l} x \xrightarrow{S} y \xrightarrow{\text{Delay } \tau} y(t - \tau) \\ x \xrightarrow{\text{Delay } \tau} x(t - \tau) \xrightarrow{S} y(t - \tau) \end{array}$$

} Gives same output

Example 1

Is $S_{ex}(t) = [x(t)]^2$ a time-invariant system?

$$\begin{array}{l} x(t) \xrightarrow{\text{System}} [x(t)]^2 \xrightarrow{\text{Delay}} [x(t - \tau)]^2 \\ x(t) \xrightarrow{\text{Delay}} x(t - \tau) \xrightarrow{\text{System}} [x(t - \tau)]^2 \end{array} \quad \therefore \text{Same, time-invariant}$$

Example 2

Is $y(t) = x(2t)$ a time-invariant system?

$$\begin{array}{l} x(t) \xrightarrow{\text{system}} x(2t) \xrightarrow{\text{Delay } \tau} x(2(t - \tau)) = x(2t - 2\tau) \\ x(t) \xrightarrow{\text{Delay } \tau} x(t - \tau) \xrightarrow{\text{System}} x(2t - \tau) \end{array}$$

$\therefore$ Since $x(2t - 2\tau) \neq x(2t - \tau)$, $y(t) = x(2t)$ is time-varying.

Linearity

A linear system preserves linear combination

$$\text{Condition ① : } S(\alpha x) = \alpha S(x)$$

$$\text{Condition ② : } S(x_1 + x_2) = S(x_1) + S(x_2)$$

$$\text{Combined condition : } S(\alpha x_1 + \beta x_2) = \alpha S(x_1) + \beta S(x_2)$$

for every x_1, x_2, α and β

Example 1

Is $S(x)(t) = [x(t)]^2$ linear?

$$\text{Let } x = \alpha x_1 + \beta x_2,$$

$$S(\alpha x_1 + \beta x_2) = [\alpha x_1 + \beta x_2]^2 \text{ --- ①}$$

$$\alpha S(x_1) + \beta S(x_2) = \alpha (x_1)^2 + \beta (x_2)^2 \text{ --- ②}$$

Since ① $\neq$ ②,
 $\Rightarrow$ Non Linear

Example 2

Is the following linear?

$$y[n] = \frac{1}{3}(x[n-1] + x[n] + x[n+1])$$

$$S(x)[n] = \frac{1}{3}(x[n-1] + x[n] + x[n+1])$$

$$\text{Let } x = \alpha x_1 + \beta x_2,$$

$$\begin{aligned} \Rightarrow S(\alpha x_1 + \beta x_2) &= \frac{1}{3}[(\alpha x_1[n-1] + \beta x_2[n-1]) + (\alpha x_1[n] + \beta x_2[n]) + (\alpha x_1[n+1] + \beta x_2[n+1])] \\ &= \alpha \frac{1}{3}[x_1[n-1] + x_1[n] + x_1[n+1]] + \beta \frac{1}{3}[x_2[n-1] + x_2[n] + x_2[n+1]] \end{aligned}$$

$$S(\alpha x_1 + \beta x_2) = \alpha S(x_1) + \beta S(x_2)$$

$\therefore$ Linear.

Linear Time-Invariant (LTI) systems

A system is LTI if it is simultaneously Linear & Time-Invariant.
Test these 2 properties separately

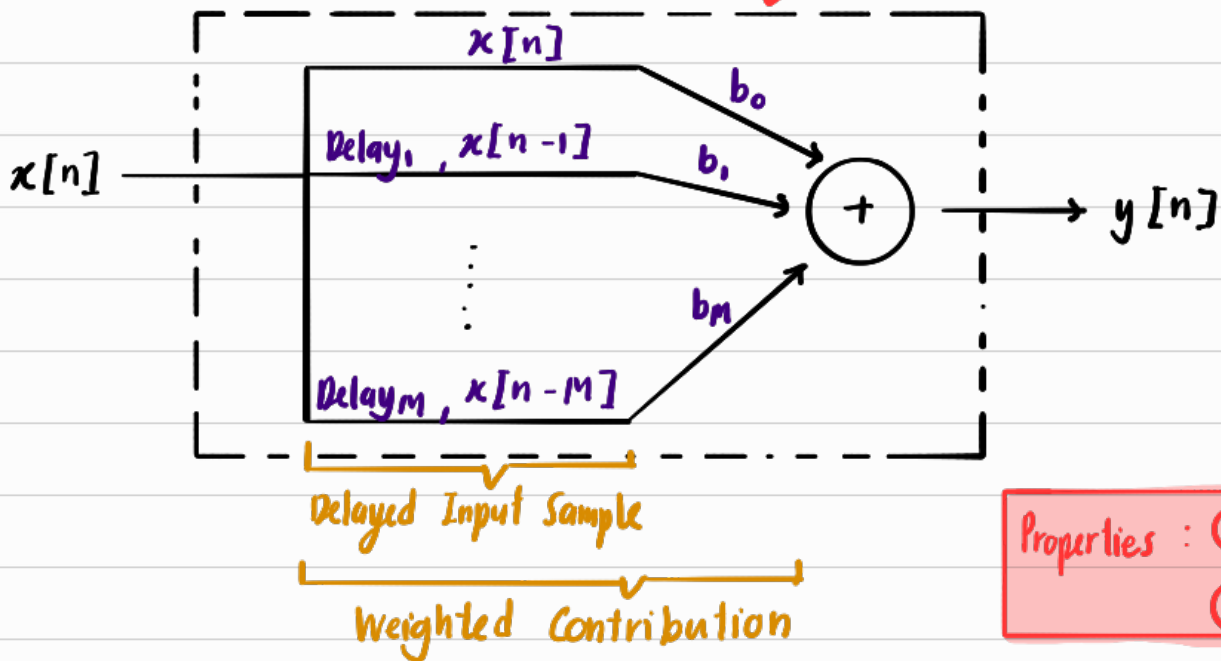
Difference Equations (DT systems) [LTI]

System Used...

- Delay
- Adders
- Gain
- Parallel Interconnection

Linear Constant-coefficient difference Eqn 1

Non-recursive causal case



Properties : ① LTI
② Causal

$$y[n] = b_0 x[n] + b_1 x[n-1] + \dots + b_M x[n-M]$$

Constants

Linear Constant-coefficient difference Eqn II

Recursive Causal Case

$$a_0 y[n] + a_1 y[n-1] + \dots + a_N y[n-N] = b_0 x[n] + b_1 x[n-1] + \dots + b_M x[n-M]$$

This is called recursive because previous output values feed into the determination of the new output

To find the output signal, $y[n]$ we need to know the previous outputs $y[n-1]$, $y[n-2]$, ..., $y[n-N]$. These are called the **initial conditions**.

Condition of initial rest

$$x[n] = 0 \text{ for all } n < n_0 ; y[n] = 0 \text{ for all } n < n_0$$

Before the input starts, the system has no pre-existing stored output response.

Only if condition of initial rests holds, the recursive causal case system will be causal and LTI

Differential Equations [CT system] [LTI]

Linear Constant-coefficient Differential Eqn

$$a_0 y + a_1 y' + \dots + a_N y^{(N)} = b_0 x + b_1 x' + \dots + b_m x^{(m)}$$

Only if condition of initial rests holds, the recursive causal case system will be causal and LTI